

AN INTRODUCTION TO WEINSTEIN/STEIN/LIOUVILLE MANIFOLDS AND SYMPLECTIC FILLINGS

CEM YALIM ÖZEL AND CAHIT ERDEM ANIL

ABSTRACT. In the present notes, we explain different types of symplectic fillings of contact manifolds, symplectic cobordisms, and Weinstein handlebody theory in relation of Lefschetz fibrations.

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1. EXACT SYMPLECTIC MANIFOLDS

Definition 1.1 (Symplectic manifold). A *Symplectic manifold* is a pair (W^{2n}, ω) where ω is a 2-form satisfying:

- closed: $d\omega = 0$

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- Non-degenerate: For each $p \in W$, the bilinear form $\omega_p : T_p W \times T_p W \rightarrow \mathbb{R}$ is non degenerate i.e

$$\omega_p(v, w) = 0 \text{ for all } w \in T_p W \Rightarrow v = 0$$

Definition 1.2 (Exact Symplectic Manifold). An *exact symplectic manifold* is a triple $(W^{2n}, \omega, \lambda)$, where:

- (W^{2n}, ω) is a symplectic manifold with ω nondegenerate, closed 2-form,
- the 1-form λ satisfies $\omega = d\lambda$.

The chosen primitive λ is called the *Liouville form*. The unique vector field Z_λ satisfying $\iota_{Z_\lambda} \omega = \lambda$ is called the *Liouville vector field*.

Remark 1.3. The Liouville form, λ is not unique: If λ is a Liouville form for ω , then so is $\lambda + df$ for any function f smooth, since d linear and $d^2 = 0$.

Remark 1.4. However the non-degeneracy of ω makes Z unique. As $0 = \iota_{Z_1} \omega - \iota_{Z_2} \omega = \iota_{Z_1 - Z_2} \omega$, as interior product is linear with respect to vector fields and by the definition of the interior product we have for every point $p \in W$ and $v \in T_p W$ $\omega_p((Z_1)_p - (Z_2)_p, v) = 0$. By the nondegeneracy $(Z_1)_p = (Z_2)_p$ for every $p \in W \Rightarrow Z_1 = Z_2$

Proposition 1.5. Let (W^{2n}, ω) be a symplectic manifold. Let λ be a 1-form, and Z be a vector field on W satisfying $\iota_Z \omega = \lambda$. Then

$$d\lambda = \omega \iff L_Z \omega = \omega$$

Proof. Equivalently, a vector field Z is a Liouville vector field (for some Liouville form λ) if and only if $L_Z \omega = \omega$

($\Rightarrow$) Let $d\lambda = \omega$, by Cartan's magic formula (for any vector field Z and k -form ω , $L_Z \omega = d(\iota_Z \omega) + \iota_Z(d\omega)$),

$$L_Z \omega = d(\iota_Z \omega) + \iota_Z(d\omega) = d(\iota_Z \omega) + \iota_Z(\omega) = d(\iota_Z \omega) + \lambda = d\lambda = \omega$$

by closedness and linearity of inner product.

($\Leftarrow$) Let $\omega = L_Z \omega$, then

$$\omega = L_Z \omega = d(\iota_Z \omega) + \iota_Z(d\omega) = d(\iota_Z \omega) + 0 = d(\lambda)$$

again by the Cartan's magic formula and the assumption $\iota_Z \omega = \lambda$. □

Remark 1.6. Because ω is nondegenerate, the map $: TW \rightarrow T^*W$, $Z \mapsto \iota_Z \omega = \omega(Z, -)$ is an isomorphic correspondence of vector fields and 1-forms on a symplectic manifold (W^{2n}, ω) , using the above proposition, one may prescribe an exact symplectic manifold by defining either its Liouville vector field, or its Liouville 1-form. In light of this, (W^{2n}, ω, Z) shall denote a symplectic manifold (W^{2n}, ω) together with Liouville vector field Z satisfying $L_Z \omega = \omega$, or equivalently a symplectic manifold (W^{2n}, ω) with a Liouville 1-form λ satisfying $(W^{2n}, \omega, \lambda)$ with $d\lambda = \omega$.

Remark 1.7. By integrating the equation $L_Z\omega = \omega$, (and solving the ODE $y' = y$ with initial condition $y(0) = (Z^0)^*\omega$) we observe a dynamical/geometric interpretation of the Liouville vector field Z of an exact symplectic manifold (W^{2n}, ω, Z) . The flow¹ $Z^t : W \rightarrow W$ expands the symplectic form exponentially, that is $(Z^t)^*\omega = e^t\omega$. Consequently the volume expands, $(Z^t)^*\omega^n = e^{nt}\omega^n$. Of course in general we can not assume that the Liouville vector field Z is complete. We immediately observe from the volume expansion that if Z is complete then W must have infinite volume. In later sections we shall identify a natural class of exact manifolds with complete Liouville vector fields.

Example 1.8. (Radial Liouville Vector Field)

Consider the standard symplectic space $(\mathbb{R}^{2n}, \omega_{std})$, where $\omega_{std} = \sum_{j=1}^n dx_i \wedge dy_j$.

A choice of Liouville form² is:

$$\lambda = \frac{1}{2} \sum_{j=1}^n (x_j dy_j - y_j dx_j)$$

with Liouville vector field,

$$Z = \frac{1}{2} \sum_{j=1}^n (x_j \frac{\partial}{\partial x_j} + y_j \frac{\partial}{\partial y_j})$$

so called the Radial vector field.

Verification of Z is Liouville vector field As Z is a vector field on $\mathbb{R}^{2n}$ the general form of Z is $\sum_{j=1}^n (a_j \frac{\partial}{\partial x_j} + b_j \frac{\partial}{\partial y_j})$ with a_j, b_j smooth functions on $\mathbb{R}^{2n}$ to $\mathbb{R}$ to be determined.

$$\iota_Z\omega = \iota_Z(\sum_{k=1}^n dx_k \wedge dy_k) = \sum_{k=1}^n \iota_Z(dx_k \wedge dy_k) = \sum_{k=1}^n (a_k dy_k - b_k dx_k)$$

Matching this $\iota_Z\omega$ with our chosen λ , requires $a_j = \frac{x_j}{2}, b_j = \frac{y_j}{2}$, giving the radial vector field:

$$Z = \frac{1}{2} \sum_{j=1}^n (x_j \frac{\partial}{\partial x_j} + y_j \frac{\partial}{\partial y_j})$$

The Flow (Z^t): To find the flow, we solve $x'_j = \frac{1}{2}x_j, y'_j = \frac{1}{2}y_j$ and obtain the Liouville flow:

$$Z^t(x, y) = (e^{t/2}x, e^{t/2}y)$$

Note that the flow expands the symplectic form exponentially as explained above i.e. $(Z^t)^*\omega = \sum_{j=1}^n d(e^{t/2}x_j) \wedge d(e^{t/2}y_j) = e^t \sum_{j=1}^n dx_j \wedge dy_j = e^t\omega$.

Definition 1.9. Let (M^{2n}, ω) and (N^{2n}, ω') be symplectic manifolds. A diffeomorphism $\varphi : M^{2n} \rightarrow N^{2n}$ is called a **symplectomorphism** if $\varphi^*\omega' = \omega$ i.e diffeomorphism preserves symplectic forms.

Remark 1.10 (Properties of Pullback). For $\varphi : M \rightarrow N$ smooth map.

¹We denote the time- t map of the flow generated by a vector field Z by Z^t .

²Exactness ($d\lambda = \omega$), immediately comes from the graded Leibniz rule, $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for k is the degree of the differential form α .

- **Respects wedge product** For α, β k-forms on N ,

$$\varphi^*(\alpha \wedge \beta) = \varphi^*\alpha \wedge \varphi^*\beta$$

- **Respects exterior derivative** $\varphi^*(d\alpha) = d(\varphi^*\alpha)$

Definition 1.11 (Exact Symplectomorphism). Let $(W^{2n}, \omega, \lambda)$ and $(W'^{2n}, \omega', \lambda')$ be exact symplectic manifolds. A diffeomorphism $\varphi : W \rightarrow W'$ is

- an *exact symplectomorphism* if $\varphi^*\lambda' = \lambda + df$ for some smooth function $f : W \rightarrow \mathbb{R}$.
- In the special case $f = 0$, i.e $\varphi^*\lambda' = \lambda$, the diffeomorphism φ is called a *strict exact symplectomorphism*.

The two exact symplectic manifolds are called *exact symplectomorphic*, and *strictly exact symplectomorphic*, respectively.

Remark 1.12. Exact symplectomorphisms are symplectomorphisms. Explicitly,

$$\varphi^*\omega' = \varphi^*(d\lambda') = d(\varphi^*\lambda') = d(\lambda + df) = d\lambda + d^2f = \omega$$

A general symplectomorphism $\varphi : W^{2n} \rightarrow W'^{2n}$ of exact symplectomorphic manifolds $(W^{2n}, \omega, \lambda)$ and $(W'^{2n}, \omega', \lambda')$ necessarily satisfies $\varphi^*\lambda' = \lambda + \alpha$ with $d\alpha = 0$. In particular, if the deRham cohomology group in degree 1 is trivial, $H_{dR}^1(W^{2n}) = 0$, φ is automatically exact.

Example 1.13. A non-example (Symplectomorphic $\nRightarrow$ Exact symplectomorphic):

Consider the standart Euclidean plane $\mathbb{R}^2$ with polar coordinates (r, θ) ³ and the standart symplectic form $\omega = 2\pi r dr \wedge d\theta$. To simplify notation, denote by $p = \pi r^2$, then $\omega = dp \wedge d\theta$. Consider the two symplectic annuli with unit area given as follows:

- $A_1 = \{(p, \theta) \in [1, 2] \times S^1\}$
- $A_2 = \{(p, \theta) \in [2, 3] \times S^1\}$

Consider the diffeomorphism $\varphi : A_1 \rightarrow A_2$ such that $\varphi(p, \theta) = (p + 1, \theta)$ which is also a symplectomorphism as it preserves the symplectic form

$$\varphi^*\omega = \varphi^*(dp \wedge d\theta) = d(p + 1) \wedge d\theta = dp \wedge d\theta = \omega.$$

If φ were an exact symplectomorphism then the pullback of Liouville form would differ by df for some smooth $f : A_1 \rightarrow \mathbb{R}$, i.e $\varphi^*\lambda = \lambda + df$. Integrating over boundary component C yields

$$\int_{\varphi(C)} \lambda = \int_C \varphi^*\lambda = \int_C \lambda + \int_C df = \int_C \lambda.$$

So exactness of our map φ would force $\int_{\varphi(C)} \lambda = \int_C \lambda$ for any boundary component C .

Geometrically, this means that exact symplectomorphisms preserve the flux of the Liouville vector field on mapped loops. Explicitly, the radial Liouville vector field on $\mathbb{R}^{2n}$ is $Z = \frac{1}{2}r \frac{\partial}{\partial r}$.

³Here θ is valued in $S^1 = \mathbb{R}/\mathbb{Z}$. That is, it is $1/2\pi$ the usual angular parameter. This is the angle parameter of the action-angle coordinates associated to the 1-periodic Hamiltonian given by πr^2 .

By Stokes' theorem, flux of Z across any C is equal to the total symplectic area enclosed by that curve:

$$Flux_C(Z) = \int_C \iota_Z \omega = \int_C \lambda = \int_{enclosed} \omega = \text{Enclosed area}$$

Now since $\lambda = pd\theta$, integrating Liouville form over any boundary curve gives the symplectic area enclosed by that curve. But,

- On the C_{in} of A_1 , enclosed area is 1 so, $\int_{C_{in}} \lambda = 1$
- The image $\varphi(C_{in})$ enclosed area is 2, so $\int_{\varphi(C_{in})} \lambda = 2 \neq 1$.

Which violates the exactness condition.

1.1. Symplectizations. [MS, p. 137] [CE, p. 128]

Definition 1.14 (Symplectization). Let (M^{2n+1}, ξ) be a co-oriented contact manifold with contact form α . We denote by $\mathcal{SM}$ the symplectization of (M^{2n+1}, ξ) , the exact symplectic manifold $(\mathbb{R} \times M^{2n+1}, d(e^s \pi^* \alpha))$ where $s : \mathbb{R} \times M^{2n+1} \rightarrow \mathbb{R}$, is the projection and $\pi : \mathbb{R} \times M^{2n+1} \rightarrow M^{2n+1}$ is the projection onto the second factor.

Remark 1.15. If $\dim M = 2n - 1$ then $\omega^n = (e^t(dt \wedge \alpha + d\alpha))^n = ne^{nt} dt \wedge \alpha \wedge (d\alpha)^{n-1} \neq 0$ so the form $\omega = (de^s \alpha)$ is a symplectic form. ⁴Although the construction apriori depends on the chosen contact form α , it does not as the following proposition shows. We shall denote this exact symplectic manifold by $(\mathcal{SM}, \omega_\xi = d\lambda_\xi)$.

Proposition 1.16 (Symplectization is independent of the choice of the contact form). *Let (M, ξ) be a co-oriented contact manifold. Let α , and α' be two contact forms. The exact symplectic manifolds $(\mathbb{R} \times M, d(e^s \pi^* \alpha))$ and $(\mathbb{R} \times M, d(e^s \pi^* \alpha'))$ are strictly exact symplectomorphic.*

Proof. As α, α' differ by a scaling factor⁵, for some smooth $h : M \rightarrow \mathbb{R}$ we can write $\alpha = e^h \alpha'$.

On the symplectization $SM = \mathbb{R} \times M$, let the two Liouville forms be $\lambda = e^s \pi^* \alpha$ and $\lambda' = e^s \pi^* \alpha'$. Using the relation we just found, $\lambda' = e^s \pi^* (e^h \alpha) = e^{s+h \circ \pi} \pi^* \alpha$.

⁴By abuse of notation, we shall often denote $\pi^* \alpha$ simply as α .

⁵Let ξ be cooriented contact structure on M . Since α and α' both contact forms defining the same co-oriented contact structure ξ . Their kernels are the same ($\ker(\alpha) = \ker(\alpha') = \xi$). This implies they only differ by a scaling factor (non-vanishing smooth), $f : M \rightarrow \mathbb{R}$. As both 1-form smooth, f smooth and as they both define the same co-oriented contact structure they induce the same orientation on its normal bundle TM/ξ implies f strictly positive.

Let $\phi : \mathbb{R} \times M \rightarrow \mathbb{R} \times M$ be given by $\phi(s, x) = (s - h(x), x)$. Now we compute the pullback of λ' under ϕ :

$$\begin{aligned}\phi^* \lambda' &= \phi^*(e^s \pi^* \alpha') \\ &= \phi^*(e^s) \phi^*(\pi^* \alpha') \\ &= e^{s - \pi^* h} (\pi \circ \phi)^* \alpha' \\ &= e^{s - \pi^* h} \pi^* \alpha' \\ &= e^s \pi^* \alpha = \lambda.\end{aligned}$$

□

1.2. Hypersurface of Contact Type. [MS, p. 142] [CE, p. 128]

Definition 1.17 (Contact Type Hypersurface). Let (W^{2n}, ω) be a symplectic manifold, and $M^{2n-1} \subset W$ a co-oriented hypersurface (codimension 1 submanifold). We say that M is a hypersurface of contact type if there exists a 1-form λ defined on a neighborhood of M such that $d\lambda = \omega$, and the Liouville vector field Z (defined by $\iota_Z \omega = \lambda$) is positively transverse to M . We shall say that $(W, \omega, M, \lambda, Z)$ is a contact situation.⁶

Example 1.18 (Star-Shaped Hypersurface). A hypersurface $M \subset \mathbb{R}^{2n}$ is star-shaped (with respect to origin) if every ray from the origin intersects M transversely at exactly one point. Let $M \subset \mathbb{R}^{2n}$ be a star shaped hypersurface, co-oriented by its outer normal and Z is positively transverse to M . Hence, $(\mathbb{R}^{2n}, \omega, M, Z)$ is a contact situation, where Z is the radial Liouville vector field defined in Example ?? as $Z = \frac{1}{2} \sum_{j=1}^n (x_j \partial_{x_j} + y_j \partial_{y_j})$.

Verification of contact structure: As Z positively transverse to M , (M, Z) hypersurface of contact type and contact form is $\alpha = \lambda|_{TM}$.

Examples of Star-Shaped Hypersurfaces:

- Spheres: The sphere $S^{2n-1} = \{(x, y) \in \mathbb{R}^{2n} : \sum_{j=1}^n (x_j^2 + y_j^2) = 1\} \cap \{|z|^2 = 1\} \subset \mathbb{R}^{2n}$ is starshaped. The standart contact form is $\alpha = \frac{1}{2} \sum_{j=1}^n (x_j dy_j - y_j dx_j)|_{TS^{2n-1}}$ and the contact structure $\xi = \ker(\alpha)$ is the standart contact structure son S^{2n-1}
- Ellipsoids: Let a_i be positive constants, the ellipsoid $E(a_1, \dots, a_n) = \{\sum_{j=1}^n \frac{x_j^2 + y_j^2}{a_j} = 1\}$ is starshaped with respect to origin (every ray from the origin intersects the ellipsoid once, transversely). Therefore $E(a_1, \dots, a_n)$ is of contact type, with contact form $\alpha = \lambda|_E$.
- Boundaries of Starshaped domains: If $\Omega \subset \mathbb{R}^{2n}$ is a bounded domain with origin and smooth boundary, $M = \partial\Omega$. If Ω starshaped with respect to origin (i.e $0 \in \Omega$, and for all $p \in \Omega$, $[0, p] \subset \text{cl}(\Omega)$ and meets M only at p), then M is hypersurface of contact type!
- Perturbed spheres: let f be a smooth positive function on S^{2n-1} , define $M_f = \{f(\theta) \cdot \theta : \theta \in S^{2n-1}\}$. Then M_f is starshaped hence of contact type! This is the general definition of starshaped.

⁶When it is clear, we might omit some of the terms from the quadruple, e.g (W, M, λ) or (W, ω, M, Z) .

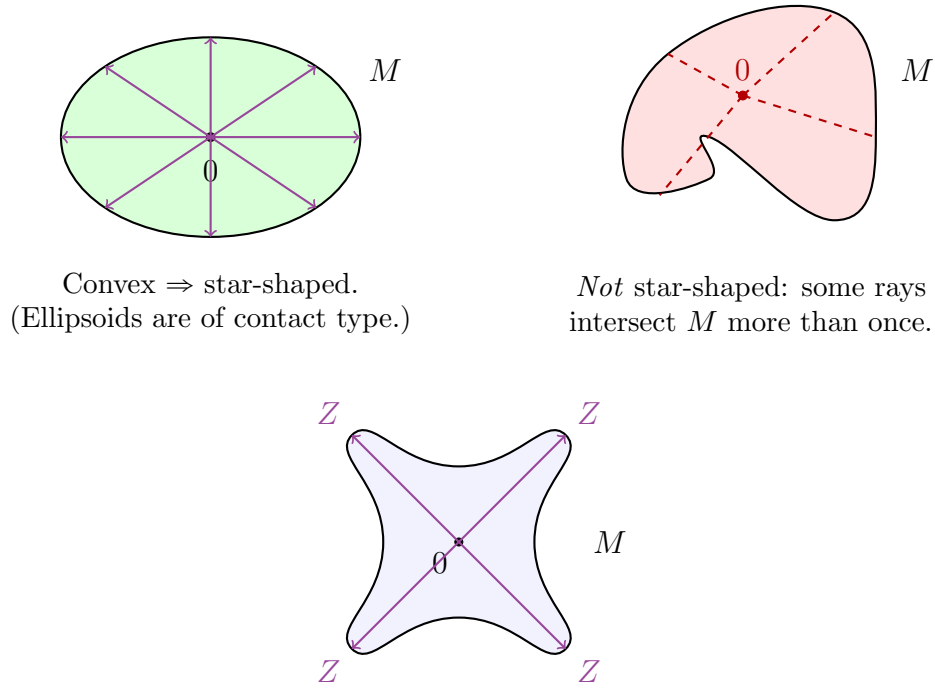


FIGURE 1. A starshaped hypersurface $M \subset \mathbb{R}^{2n}$ with the radial Liouville vector field Z positively transverse to M at every point, making $(M, \lambda|_{TM})$ a contact manifold.

Remark 1.19. Rabinowitz proved [Rab] that star-shaped hypersurfaces have closed characteristics (a closed leaf of the characteristic foliation). Weinstein, in search of a symplectomorphism invariant characterization of being star shaped, introduced the notion of contact type hypersurfaces [Wei] and conjectured that they all have closed characteristics. A slight generalization of this conjecture is now known as the Weinstein conjecture.

Example 1.20. Consider the symplectization $\mathcal{S}M \cong \mathbb{R} \times M$ of a contact manifold (M, ξ) . Then the surfaces $\Gamma_f = \{(f(x), x) : x \in M\}$, where $f : M \rightarrow \mathbb{R}$ is a smooth function, are of contact type. More precisely $(\mathcal{S}M, \Gamma_f, \lambda_\xi)$ is a contact situation. Note that $\hat{f} = f \times id : M \rightarrow \mathbb{R} \times M$ pulls back $\hat{f}^* \lambda_\xi = e^f \alpha$.

Proposition 1.21. Let (W^{2n}, ω) be symplectic, and $M \subset W$ a hypersurface of contact type. Let Z be a particular Liouville vector field which is positively transverse to M with corresponding Liouville form λ .

- (1) The restriction $\alpha = \lambda|_{TM}$ is a positive contact form on M .
- (2) Let $(W, \omega, M, \lambda', Z')$ be another contact situation. Let $\alpha' = \lambda'|_{TM}$. Assume furthermore that M is closed. Then (M, α) and (M, α') are contactomorphic by a diffeomorphism isotopic to the identity. Assume furthermore that $\alpha - \alpha'$ is exact, then the two contact manifolds (M, α) and (M, α') are strictly contactomorphic by a strict contactomorphism which is isotopic to the identity.

- (3) *There is an open neighborhood of U of $0 \times M$ in $(\mathbb{R} \times M, d(e^s \alpha))$ for which $\Phi : U \rightarrow W$ defined by $\Phi : (s, x) \mapsto Z^s(x)$ is a symplectic embedding. When M is compact, we may choose U to be of the form $(-\varepsilon, \varepsilon) \times M$.*
- (4) *Let $H : W \rightarrow \mathbb{R}$ be a function which agrees with $s \circ \Phi^{-1}$ near M . In particular M consists of regular points in $H^{-1}(0)$. Then we have the equivalence of the Reeb vector field R_α and the Hamiltonian vector field X_H on M .*

Proof. (1) Let $p \in M$, and $v_1, \dots, v_{2n-1} \in T_p M$ be a positively oriented basis of $T_p M$. Then $Z(p), v_1, \dots, v_{2n-1}$ is a positively oriented basis of $T_p W$. We compute

$$\begin{aligned} \alpha \wedge (d\alpha)^{n-1}(v_1, \dots, v_{2n-1}) &= (\lambda \wedge (d\lambda)^{n-1})|_{T_p M}(v_1, \dots, v_{2n-1}) \\ &= (\iota_Z \omega \wedge (\omega)^{n-1})|_{T_p M}(v_1, \dots, v_{2n-1}) \\ &= \omega^n(Z, v_1, \dots, v_{2n-1}) > 0. \end{aligned}$$

Hence α is a contact form.

- (2) Denote by $\lambda_t = (1-t)\lambda + t\lambda'$ for $t \in [0, 1]$, and $Z_t = (1-t)Z + tZ'$. For each $t \in [0, 1]$, $(W, \omega, M, \lambda_t, Z_t)$ is another contact situation. Denote by $\alpha_t = (\lambda_t)|_{TM}$, and $\xi_t = \ker(\alpha_t)$. By (1), each ξ_t is a contact structure. One uses the Moser argument in the contact setting (also called Gray stability) to generate an isotopy that maps ξ_0 to ξ_1 . Note that this isotopy in general fails to be a strict contactomorphism.

For the second part, assume that $\alpha - \alpha' = dH$ for some function $H : M \rightarrow \mathbb{R}$. Denote by R_t the Reeb vector field of α_t . Let $X_t = HR_t$. Let φ_t denote the isotopy generated by X_t . Then we compute

$$\begin{aligned} \frac{d}{dt} \varphi_t^* \alpha_t &= \varphi_t^* \left(\frac{d}{dt} \alpha_t + L_{X_t} \alpha_t \right) \\ &= \varphi_t^* \left(\frac{d}{dt} \alpha_t + dH \right) = 0. \end{aligned}$$

- (3) Let $p \in M$. For $\varepsilon = \varepsilon(p) > 0$ sufficiently small, there is some neighborhood $V \subset M$ of p such that $\Phi : (s, x) \mapsto Z^s(x)$ is a smooth embedding for $(s, x) \in (-\varepsilon, \varepsilon) \times V$. To see that it is also an exact symplectomorphism onto its image, we compute

$$(\Phi^* \lambda)_{(s,x)}(\partial_s) = \lambda(Z(Z^s(x))) = 0.$$

Hence $\Phi^* \lambda$ is determined by its restriction on the tangent spaces $T(\{s\} \times V)$. To be extra pedantic, denote by $p_s : \{s\} \times V \rightarrow V$ the map $p_s(s, x) = x$. Let $s \in (-\varepsilon, \varepsilon)$, then we have

$$\begin{aligned} (\Phi^* \lambda)|_{T(\{s\} \times V)} &= (\Phi|_{\{s\} \times V})^* (\lambda|_{T(\Phi(\{s\} \times V))}) \\ &= p_s^* ((Z^s)^* \lambda)|_{TV} \\ &= p_s^* e^s \lambda|_{TV} \\ &= (e^s \pi^* \alpha)|_{T\{s\} \times V}. \end{aligned}$$

Since the canonical Liouville form $e^s \pi^* \alpha$ also vanishes on the vector ∂_s , it follows that $\Phi^* \lambda = e^s \pi^* \alpha$.

The union of such neighborhoods gives an open $U \subset \mathbb{R} \times M$ for which the map $(s, x) \mapsto Z^s(x)$ is an exact symplectic embedding.

- (4) Since Hamiltonian vector fields behave naturally under symplectomorphism, in light of (3), we may suppose that $(W, \omega, M, \lambda, Z) = (\mathbb{R} \times M, d(e^s \pi^* \alpha), \{0\}M, e^s \alpha, \partial_s), \Phi = id$, and $H = s$. We then simply compute to see that ⁷

$$\alpha(X_H) = (\iota_{\partial_s} \omega)(X_H) = \omega(\partial_s, X_H) = \partial_s H = 1.$$

$$\iota_{X_H} d\alpha = (\iota_{X_H} \omega)|_{\{0\} \times M} = -dH|_{\{0\} \times M} = 0.$$

The above two properties characterize the Reeb vector field, hence $X_H = R_\alpha$.

□

2. LIOUVILLE MANIFOLDS

2.1. Convex Boundary. [MS, p. 143]

Definition 2.1 (ω -convex). Let (W, ω) be a symplectic manifold with boundary. Let M be a union of connected components of the boundary ∂W . If M is of contact type with the outwards co-orientation, then we say that M is convex (or ω -convex, if we have to specify), if M is of contact type with the inwards co-orientation, we say that M is concave.

Example 2.2. The standard ball $B^{2n} \subset \mathbb{R}^{2n}$ has convex boundary. Similarly, for any contact (M, ξ) the negative cylindrical end $(\mathbb{R}_{\leq 0} \times M, d(e^s \alpha))$ has convex boundary, whereas the positive cylindrical end $(\mathbb{R}_{\geq 0} \times M, d(e^s \alpha))$ has concave boundary.

Proposition 2.3. *Let (W, ω) and (W', ω') be symplectic manifolds with boundary. Let $M \subset W$ be a convex boundary component with induced contact structure ξ , and $M' \subset W'$ be a concave boundary component with induced contact structure ξ' . Let $\varphi : M \rightarrow M'$ be a contactomorphism. We may glue W and W' using φ to obtain a symplectic manifold (WW, Ω) where $WW = W \cup_\varphi W'$, and $(W^{(\prime)}, \Omega|_{W^{(\prime)}})$ is deformation equivalent to $(W^{(\prime)}, \omega^{(\prime)})$ along symplectic structures keeping $M^{(\prime)}$ convex (concave).*

If we assume that there exists Liouville forms $\lambda^{(\prime)}$ near $M^{(\prime)}$ such that $\alpha^{(\prime)} = \lambda^{(\prime)}|_{M^{(\prime)}}$ are contact forms, and that φ is a strict contactomorphism $\varphi : (M, \alpha) \rightarrow (M', \alpha')$, then we may suppose that $(W^{(\prime)}, \Omega|_{W^{(\prime)}}) = (W^{(\prime)}, \omega^{(\prime)})$.

2.2. Symplectic Cobordism.

Definition 2.4 ((Strong) Symplectic Cobordism). Let $(W, \partial W = M_- \amalg M_+, \omega)$ be a compact symplectic manifold with boundary such that M_- is concave, and M_+ is convex, with induced contact structures $(M_\pm, \xi_\pm)$. Then we say that (W, ω) is a (strong) symplectic cobordism from (M_-, ξ_-) to (M_+, ξ_+) . In the case $M_- = \emptyset$, we say that (W, ω) is a strong filling of (M_+, ξ_+) .

If for a given contact manifold (M, ξ) one can find a strong filling $(W, \omega, \partial W = M)$, we say that (M, ξ) is strongly fillable.

⁷The sign convention for the Hamiltonian may not agree with the other weeks'.

Example 2.5. The standard symplectic ball $B^{2n} \subset \mathbb{R}^{2n}$ fills the standard contact sphere. This follows from ..

Remark 2.6. The compactness condition in Definition 2.4 is crucial. Indeed, certain closed contact manifolds are not fillable, so it is a non-trivial property of the contact manifold. However there are always 'non-compact fillings', namely one can consider the negative cylindrical end of the symplectization.

Remark 2.7. The direction matters.

Remark 2.8. If we forget about the extra structures, it is an oriented cobordism.

Proposition 2.9. *One may glue symplectic cobordisms.*

2.3. Liouville Cobordisms.

Definition 2.10. A Liouville cobordism is of the form $(W, \omega, \partial W = M_- \amalg M_+, \omega = d\lambda)$, where $(W, \omega, M_- \amalg M_+)$ is a symplectic cobordism, and λ is a primitive of ω such that the associated Liouville vector field Z is (inwards) outwards transverse to $(M_-) M_+$. In which case we say that (W, ω) is a Liouville ⁸ cobordism from (M_-, ξ_-) to (M_+, ξ_+) . In case $M_- = \emptyset$, we say that (W, ω) is a Liouville filling of (M_+, ξ_+) , and call (W, ω) a Liouville domain.

If for a given contact manifold (M, ξ) one can find a strong filling $(W, \omega, \partial W = M)$, we say that (M, ξ) is Liouville fillable.

Example 2.11 (A strong filling which can not be made exact). The standard ball B^{2n} fills the standard sphere S^{2n-1} . Blowing up a small ball in the interior (for discussion on constructing symplectic forms on blow ups see [MS] Section 7.1) yields a non-exact symplectic manifold.⁹ Hence the ball blown up once yields a non exact symplectic manifold, meaning this strong filling can not be made exact.

Example 2.12. Ghiggini [Ghi] produces examples of contact 3-manifolds which are strongly fillable and yet not Liouville fillable.

2.4. Weinstein Cobordism. [MS, p. 332]

Definition 2.13. A Weinstein cobordism is of the form $(W, \omega, \partial W = M_- \amalg M_+, \omega = d\lambda, \phi)$, where $(W, \omega, \partial W = M_- \amalg M_+, \omega = d\lambda)$, is a Liouville cobordism, and $\phi : W \rightarrow \mathbb{R}$ is a Morse function which is constant and regular ($d\phi$ does not vanish along the boundary) on the boundary components, and for which the Liouville vector field Z is gradient-like for ϕ ,

⁸Also called exact

⁹Recall that a symplectic blow up of a symplectic manifold at a point is removing a small Darboux ball B^{2n} and collapsing its boundary S^{2n-1} . The resulting manifold admits a new symplectic form which agrees with the old one outside the collapsed region, and the new submanifold so called *Exceptional Divisor* (say E) has $\int_E \omega_{new} > 0$. Now if new form were exact, for some Liouville form then for the new closed submanifold E would force $\int_E \omega_{new} = \int_E d\lambda = \int_{\partial E} \lambda = 0$ since $\partial E = \emptyset$. Contradiction.

viz there is some auxilliary Riemannian metric for which $Z\phi \geq \varepsilon(|d\phi|^2 + |Z|^2)$ is satisfied. In which case we say that (W, ω) is a Weinstein cobordism from (M_-, ξ_-) to (M_+, ξ_+) . In case $M_- = \emptyset$, we say that (W, ω) is a Weinstein filling of (M_+, ξ_+) , and call $(w, \omega, \lambda, \phi)$ a Weinstein domain.

If for a given contact manifold (M, ξ) one can find a strong filling $(W, \omega, \partial W = M)$, we say that (M, ξ) is Weinstein fillable.

Remark 2.14. It is common to consider completions of Weinstein cobordisms/domains, namely attaching cylindrical ends of the contact boundary components and extending the functions ϕ and the Liouville vector fields in the natural way. It is common in the literature to work with non-compact and even non-finite type Weinstein manifolds (e.g Cielibak-Eliahsberg).

Example 2.15 (Cotangent Bundles). I shall do this carefully at some point in time.

Proposition 2.16. *Let $(W, \omega, \partial W = M_+ = M, \lambda, \phi)$ be a Weinstein domain. The skeleton $\text{Skel}(W, \lambda)$ is the union of isotropic submanifolds.*

Corollary 2.17. *Let $(W^{2n}, \omega, \partial W = M_+ = M, \lambda, \phi)$ be a Weinstein domain.*

- (1) *The index of a critical point of ϕ is at most n .*
- (2) *The $2n$ -dimensional manifold W has the homotopy type of an n -dimensional CW-complex. In particular, $H_i(W) = 0$ for $i > n$.*
- (3) *The maps of homotopy groups $\iota_* : \pi_i(\partial W) \rightarrow \pi_i(W)$ induced from the inclusion $\iota : \partial W \rightarrow W$ are surjective for $i \leq n - 1$, and also injective for $i \leq n - 2$. Ditto for the maps $H_i(\partial W) \rightarrow H_i(W)$.*

Example 2.18 (A Liouville manifold which cannot be made Weinstein). McDuff's warped cotangent bundle example.

2.5. On Deformation.

3. WEINSTEIN HANDLEBODIES

Definition 3.1. An elementary Weinstein cobordism is a Weinstein cobordism $(W, \omega, \partial W = M_- \amalg M_+, \lambda, \phi)$ such that ϕ has a single critical point.

Proposition 3.2. *Elementary cobordisms are Weinstein handle attachments. (will be reworded)*

The rest of the section will be occupied with Weinstein's proof of the above fact. ¹⁰

¹⁰Caution for the reader: Weinstein adopts the opposite sign convention for Liouville vector fields, and also calls Weinstein manifolds Stein manifolds, partly because Stein manifolds in complex geometry have analogous properties and partly because it would be extremely egotistical to call something by your own name. In fact later work of Cielibak and Eliahsberg prove the (deformation) equivalence of the two structures.

3.1. Standard Form of Isotropics. Let $(W^{2n}, \omega, \lambda, Z_\lambda, M^{2n-1} \subset W^{2n}, \xi = \ker \lambda|_{TM}, Y^{k-1} \subset M^{2n-1})$ be the following: (W, ω, λ) is a Liouville manifold; $M \subset W$ is a hypersurface of contact type with induced contact structure ξ . $Y \subset M$ is an isotropic submanifold. Following Weinstein, we shall call such a septuple an isotropic setup.

Remark 3.3. The isotropic setup is specified more economically by (W, λ, M, Y) . We shall often specify our isotropic setup in this more efficient fashion.

Definition 3.4. We define the symplectic vector bundle $CSN(M, Y)$ by

$$(1) \quad SN(W, \lambda, M, Y) = (TY)^\omega \cap \xi / TY.$$

We call this bundle the symplectic subnormal bundle of Y with respect to $(W, \omega, \lambda, M, \xi, Y)$.

We denote by $CSN(M, \xi, L)$ the conformal symplectic vector bundle

$$(2) \quad CSN(M, \xi, Y)TY^\omega / TY$$

where TY^ω denotes the symplectic orthogonal complement¹¹ of TY inside ξ . We call this the conformal symplectic subnormal bundle of the isotropic Y in M, ξ .

Remark 3.5. Observe that $CSN(M, \xi, Y)$ can be defined solely by using the contact structure ξ . However it is also clearly the relaxation¹² of $SN(W, \lambda, M, Y)$.

Remark 3.6. Let $\dim M = 2n - 1$, and $\dim Y = k - 1$ where $1 \leq k \leq n$, then the rank of the bundle $CSN(M, \xi, Y)$ equals $2(n - k)$.

Proposition 3.7. *Given a symplectic vector bundle E over a smooth manifold Y , there is a functorial construction (W, λ, M, Y) such that $SN(W, \lambda, M, Y) = E$.*

If $F \rightarrow Y$ is a conformal symplectic vector bundle, we may reduce its structure group to obtain a symplectic vector bundle E , then we obtain (W, λ, M, Y) from the above paragraph. Then the resulting isotropic embedding $Y \subset M$ has associated conformal symplectic subnormal bundle $CSN(M, Y) = F$. This construction is functorial as well.

3.1.1. Framing Isotropics. Let (W, λ, M, ξ, Y) be an isotropic setup. We have the natural decomposition of the normal bundle $\nu(Y \subset W) = TW|_Y / TY$ as the direct sum of vector bundles

$$(3) \quad \nu(Y \subset W) = \nu(M \subset W) \oplus TM / \xi \oplus \xi / (TY^\omega \cap \xi) \oplus SN(W, \lambda, M, Y)$$

We have canonical isomorphisms¹³

$$\begin{aligned} \mathbb{R} &\cong \nu(M \subset W), \\ \mathbb{R} &\cong \nu(TM / \xi), \\ T^*Y &\cong \xi / (TY^\omega \cap \xi). \end{aligned}$$

¹¹Recall that a cooriented contact distribution inherits a conformal symplectic structure given by the restriction of $d\alpha$.

¹²not standard terminology, too lazy to write it properly

¹³By abuse of notation, we denote the trivial vector bundle of rank k over a manifold by $\mathbb{R}^k$, in cases where the base manifold is (hopefully) clear.

The first two trivializations are given by the non-vanishing sections Z_λ , and R_α , $\alpha = \lambda|_{TM}$, respectively. The third follows from the exact sequence

$$0 \rightarrow TY^\omega \cap \xi \rightarrow \xi \rightarrow T^*Y \rightarrow 0$$

where $\xi \rightarrow T^*Y$ is given by $X \mapsto \iota_X \omega|_{TY}$.

Now let us focus on the case when Y is diffeomorphic to a sphere S^{k-1} . Consider the standard embedding of Y into $\mathbb{R}^k$. We have the isomorphism

$$\mathbb{R}^k \cong (T^*\mathbb{R}^k)|_Y \cong \mathbb{R} \oplus T^*Y.$$

Thus (3) becomes

$$\nu(Y \subset W) \cong \mathbb{R}^{k+2} \oplus SN(W, \lambda, M, Y).$$

Hence a trivialization of $SN(W, \lambda, M, Y)$ naturally induces a framing of Y . In the case Y is Legendrian, *viz* $k = n = \frac{1}{2} \dim W$, then $SN(W, \lambda, M, Y) = 0$. Hence Y is canonically framed. Specializing even further, when Y is a Legendrian knot, that is when $n = 2$, the framing is twisted by -1 with respect to the obvious framing induced from the contact distribution. This is easily seen from the fact that the Reeb vector field does a full positive twist with respect to the canonical framing.

Example 3.8. Recall that framings of knots in S^3 are specified by integers. One may compute the canonical framing of a Legendrian knot in the standard contact S^3 using the formula:

3.1.2. *The Standard Handle.* Let $Z_k, 0 \leq k \leq n$ be the vector field defined on $\mathbb{R}^{2n}$ as follows

$$(4) \quad Z_k = \sum_{j=1}^k -x_j \partial_{x_j} + 2y_j \partial_{y_j} + \frac{1}{2} \sum_{j=k+1}^n x_j \partial_{x_j} + y_j \partial_{y_j}.$$

Let $\lambda_k = \iota_{Z_k} \omega$. Clearly $d\lambda_k = \omega$.

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DEPARTMENT OF MATHEMATICS, BOĞAZIÇI UNIVERSITY, 34342, ISTANBUL, TURKEY

Email address: yalim98@gmail.com

DEPARTMENT OF MATHEMATICS, BOĞAZIÇI UNIVERSITY, 34342, ISTANBUL, TURKEY

Email address: cahit.anil@std.bogazici.edu.tr